**Homework 36: Positive Definite Matrices**

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**BCS A, E, F, G**

**Introduction.** ***Adjoint Transformations.*** There is one final operation that is used extensively in introductory linear algebra that we have not yet generalized to vector spaces beyond $F^n$, and that is the (conjugate) transpose of a matrix. We now fill in this gap by introducing the adjoint of a linear transformation, which can be thought of as finally answering the question of why the transpose is an important operation (after all, why would we expect that swapping the rows and columns of a matrix should tell us anything useful?).

**Definition 1.** ***Adjoint Transformation.*** Suppose that $V$ and $W$ are inner product spaces and $T: V \to W$ is a linear transformation. Then a linear transformation $T^*: W \to V$ is called the ***adjoint*** of $T$ if

$$\langle T(\boldsymbol{v}), \boldsymbol{w} \rangle = \langle \boldsymbol{v}, T^*(\boldsymbol{w}) \rangle$$

for all $\boldsymbol{v} \in V$ and $\boldsymbol{w} \in W$.

For example, it is straightforward (but slightly tedious and unenlightening) to show that real matrices $A \in M_{m,n}(\mathbb{R})$ satisfy

$$\langle A(\boldsymbol{v}), \boldsymbol{w} \rangle = (A\boldsymbol{v}) \cdot \boldsymbol{w} = \boldsymbol{v} \cdot A^T \boldsymbol{w} = \langle \boldsymbol{v}, A^T(\boldsymbol{w}) \rangle$$

for all $\boldsymbol{v} \in V$ and $\boldsymbol{w} \in W$.

**Remark 1.** The ground field must be $\mathbb{R}$ or $\mathbb{C}$ in order for inner products, and thus adjoints, to make sense.

**Theorem 1.** ***Existence and Uniqueness of the Adjoint.*** Suppose that $V$ and $W$ are finite-dimensional inner product spaces with orthonormal bases $B$ and $C$, respectively. If $T: V \to W$ is a linear transformation, then there exists a unique adjoint transformation $T^*: W \to V$, and its standard matrix satisfies

$$[T^*]_{B \leftarrow C} = [T]^*{}_{C \leftarrow B}\,.$$

**Remark 2.** In infinite dimensions, some linear transformations fail to have an adjoint. However, if it exists then it is still unique.

**Definition 2.** ***Self-Adjoint Transformations.*** If $V$ is an inner product space, then a linear transformation $T: V \to V$ is called ***self-adjoint*** if $T^* = T$.

For example, a matrix in $M_n(\mathbb{R})$ is symmetric (i.e., $A = A^T$) if and only if it is a self-adjoint linear transformation on $\mathbb{R}^n$, and a matrix in $M_n(\mathbb{C})$ is Hermitian (i.e., $A = A^*$) if and only if it is a self-adjoint linear transformation on $\mathbb{C}^n$. Slightly more generally, Theorem 1 tells us that a linear transformation on a finite-dimensional inner product space $V$ is self-adjoint if and only if its standard matrix (with respect to some orthonormal basis of $V$) is symmetric or Hermitian (depending on whether the underlying field is $\mathbb{R}$ or $\mathbb{C}$).

The fact that the transposition map on $M_n$ is self-adjoint tells us that its standard matrix (with respect to an orthonormal basis, like the standard basis) is symmetric.

**Remark 3.** Strictly speaking, the transposition map on $M_{m,n}(F)$ is only self-adjoint when $m = n$, since otherwise the input and output vector spaces are different.

**Definition 3.** ***Positive (Semi)Definite Matrices.*** Suppose $A \in M_n(F)$ is self-adjoint. Then $A$ is called

**a) positive semidefinite (PSD)** if $\boldsymbol{v}^* A \boldsymbol{v} \geq 0$ for all $\boldsymbol{v} \in F^n$, and

**b) positive definite (PD)** if $\boldsymbol{v}^* A \boldsymbol{v} > 0$ for all $\boldsymbol{v} \neq \mathbf{0}$.

**Theorem 2.** ***Characterization of Positive Semidefinite Matrices.*** Suppose $A \in M_n(F)$ is self-adjoint. The following are equivalent:

**a)** $A$ is positive semidefinite,

**b)** All of the eigenvalues of $A$ are non-negative,

**c)** There is a matrix $B \in M_n(F)$ such that $A = B^*B$, and

**d)** There is a diagonal matrix $D \in M_n(\mathbb{R})$ with non-negative diagonal entries and a unitary matrix $U \in M_n(F)$ such that $A = UDU^*$.

**Problem 1.** Explicitly show that all four properties of Theorem 2 are held for the matrix

$$A = \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix}.$$

**Theorem 3.** ***Characterization of Positive Definite Matrices.*** Suppose $A \in M_n(F)$ is self-adjoint. The following are equivalent:

**a)** $A$ is positive definite,

**b)** All of the eigenvalues of $A$ are *strictly positive*,

**c)** There is an invertible matrix $B \in M_n(F)$ such that $A = B^*B$, and

**d)** There is a diagonal matrix $D \in M_n(\mathbb{R})$ with strictly positive diagonal entries and a unitary matrix $U \in M_n(F)$ such that $A = UDU^*$.

**Theorem 4.** ***Sylvester's Criterion.*** Suppose $A \in M_n$ is self-adjoint. Then $A$ is positive definite if and only if, for all $1 \leq k \leq n$, the determinant of the top-left $k \times k$ block of $A$ is strictly positive.

**Problem 2.** Find the values of $x$ and $y$ such that the following matrix is positive definite.

$$\begin{bmatrix} x & 4 \\ 4 & y \end{bmatrix}.$$

Then use *"Sylvester's Criterion"* from Theorem 4 to verify it.

**Problem 3.** Use Sylvester's criterion to show that the following matrix is positive definite.

$$A = \begin{bmatrix} 2 & -1 & i \\ -1 & 2 & 1 \\ -i & 1 & 2 \end{bmatrix}.$$

**Theorem 5.** ***Positive (Semi)Definiteness. for $2 \times 2$ Matrices.*** Suppose $A \in M_2$ is self-adjoint.

**a)** $A$ is positive semidefinite if and only if $a_{1,1}, a_{2,2}, \det(A) \geq 0$, and

**b)** $A$ is positive definite if and only if $a_{1,1} > 0, \det(A) > 0$.

**Problem 4.** Apply Theorem 4 to the matrices for *Positive (Semi)Definiteness*.

$$A = \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix}, \qquad B = \begin{bmatrix} 1 & 2 \\ 2 & 1 \end{bmatrix}.$$

**Theorem 6.** ***Positive Definite Matrices Make Inner Products.*** A function $\langle\cdot,\cdot\rangle: F^n \times F^n \to F$ is an inner product if and only if there exists a positive definite matrix $A \in M_n(F)$ such that

$$\langle \boldsymbol{v}, \boldsymbol{w} \rangle = \boldsymbol{v}^* A \boldsymbol{w}$$

for all $\boldsymbol{v}, \boldsymbol{w} \in F^n$.

**Problem 5.** ***A Weird Inner Product.***

Show that the function $\langle\cdot,\cdot\rangle: \mathbb{R}^2 \times \mathbb{R}^2 \to \mathbb{R}$ defined by

$$\langle \boldsymbol{v}, \boldsymbol{w} \rangle = v_1 w_1 + 2v_1 w_2 + 2v_2 w_1 + 5v_2 w_2$$

for all $\boldsymbol{v}, \boldsymbol{w} \in \mathbb{R}^2$ is an inner product on $\mathbb{R}^2$.

[**Hint:** You are required to attach a matrix $A$ with the inner product and show that it is a positive definite matrix. Then use Theorem 6 to show that this function is an inner product.]

**Definition 4.** ***Positive Definite Matrices.*** A symmetric matrix is positive definite iff the coefficients of the characteristic polynomial alternate in sign:

$$\det(A - \lambda I) = a_n \lambda^n - a_{n-1}\lambda^{n-1} + \cdots + (-1)^{n-1} a_1 \lambda + (-1)^n a_0\,,$$

with $a's$ either are all positive or all negative. For example, let

$$A = \begin{bmatrix} 10 & 2 & 3 \\ 2 & 1 & 1 \\ 3 & 1 & 4 \end{bmatrix}.$$

So

$$\det(A - \lambda I) = -\lambda^3 + 15\lambda^2 - 49\lambda + 17\,,$$

In which coefficients alternate in sign. So, $A$ is positive definite.

**Definition 5.** ***Positive Semi-Definite Matrices.*** The matrix $A$ is positive semidefinite iff the coefficients of the characteristic polynomial alternate in sign down to the first zero coefficient after which all coefficients are zero. Let

$$A = \begin{bmatrix} 1 & 2 & -2 \\ 2 & 4 & -4 \\ -2 & -4 & 4 \end{bmatrix}.$$

So

$$\det(A - \lambda I) = -\lambda^3 + 9\lambda^2 + 0 \cdot \lambda + 0 = -\lambda^3 + 9\lambda^2\,,$$

The first zero coefficient is the coefficient of $\lambda$ and upto this coefficient signs alternate and after that other coefficients are zero. Hence $A$ is positive semidefinite.

**Definition 6.** ***Negative Definite Matrices.*** The matrix $A$ is negative definite iff all coefficients of the characteristic polynomial has the same sign. Let

$$A = \begin{bmatrix} -1 & 3 & 0 \\ 3 & -10 & 1 \\ 0 & 1 & -4 \end{bmatrix}.$$

So

$$\det(A - \lambda I) = -\lambda^3 - 15\lambda^2 - 62\lambda - 75\,.$$

Hence all coefficients have the same sign. So, $A$ is negative definite.

**Definition 7.** ***Negative Semi-Definite Matrices.*** The matrix $A$ is negative semidefinite iff the coefficients of the characteristic polynomial have same sign upto some coefficients $a_r, r \leq n$ and after that all coefficients are zero.

**Submission Date:** November 10, 2024 (Sunday − 11:59PM)

*Good Luck*